Box 12.5

Acceleration Radiation

In 1975, Wheeler's recent student William Unruh, and independently Paul Davies at King's College, London, discovered (using the laws of quantum fields in curved spacetime) that accelerated observers just above a black hole's horizon must see the vacuum fluctuations there not as virtual pairs of particles but rather as an atmosphere of real particles, an atmosphere that Unruh called "acceleration radiation."

This startling discovery revealed that *the concept of a real particle is relative,* not absolute; that is, it depends on one's reference frame. Observers in freely falling frames who plunge through the hole's horizon see no real particles outside the horizon, only virtual ones. Observers in accelerated frames who, by their acceleration, remain always above the horizon see a plethora of real particles.

How is this possible? How can one observer claim that the horizon is surrounded by an atmosphere of real particles and the other that it is not? The answer lies in the fact that the virtual particles' vacuum fluctuational waves are not confined solely to the region above the horizon; part of each fluctuational wave is inside the horizon and part is outside.

- The freely falling observers, who plunge through the horizon, can see both parts of the vacuum fluctuational wave, the part inside the horizon and the part outside; so such observers are well aware (by their measurements) that the wave is a mere vacuum fluctuation and correspondingly that its particles are virtual, not real.
- The accelerated observers, who remain always outside the horizon, can see only the outside part of the vacuum fluctuational wave, not the inside part; and correspondingly, by their measurements they are unable to discern that the wave is a mere vacuum fluctuation accompanied by virtual particles. Seeing only a part of the fluctuational wave, they mistake it for "the real thing"—a real wave accompanied by real particles, and as a result their measurements reveal all around the horizon an atmosphere of real particles.

That this atmosphere's real particles can gradually evaporate and fly off into the external Universe (Figure 12.3c) is an indication that the viewpoint